Thesis Proposal: A Neuro-Symbolic Approach to Control Task-Oriented Dialog Systems

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Introduction

Aim: Design conversational system that deliver Heart Failure (HF) self-care
Self-care: managing salt-intake, exercise, adhering to medications [1]

Goal: Hybrid Task Oriented Dialog System (TODS) leveraging TODS and Large Language Model (LLM) capabilities.

Healthcare Dialog Systems Challenges:
- Controlling conversations to reflect system understanding
- Limited patient-centered conversational datasets
- Evaluation; Lack of appropriate metrics, Need user studies

Challenges intensified :
- African American (AA) HF patients face worst outcomes, low health literacy

Speaker	Utterance
PE	You have to ask. Um, exercise, regularly. You know, it sounds with this one to two miles you're walking on a daily basis, we're going to get you back up to that.
Patient	Okay.
PE	That's a great way to keep that going. There's no reason to stop, once we get you feeling better. Um, it used to be back in the day, maybe 20 years ago, people would say, "Well, you know, I've got to take it easy." That's not the case with heart failure. We want you to get up where you can do it. We don't want you to push yourself. . .
Patient	Right.
PE	If you're short of breath, but. . . and then, we want you to check your weight every day. Do you own a scale?

Excerpt of Patient-Educator (PE) Conversation

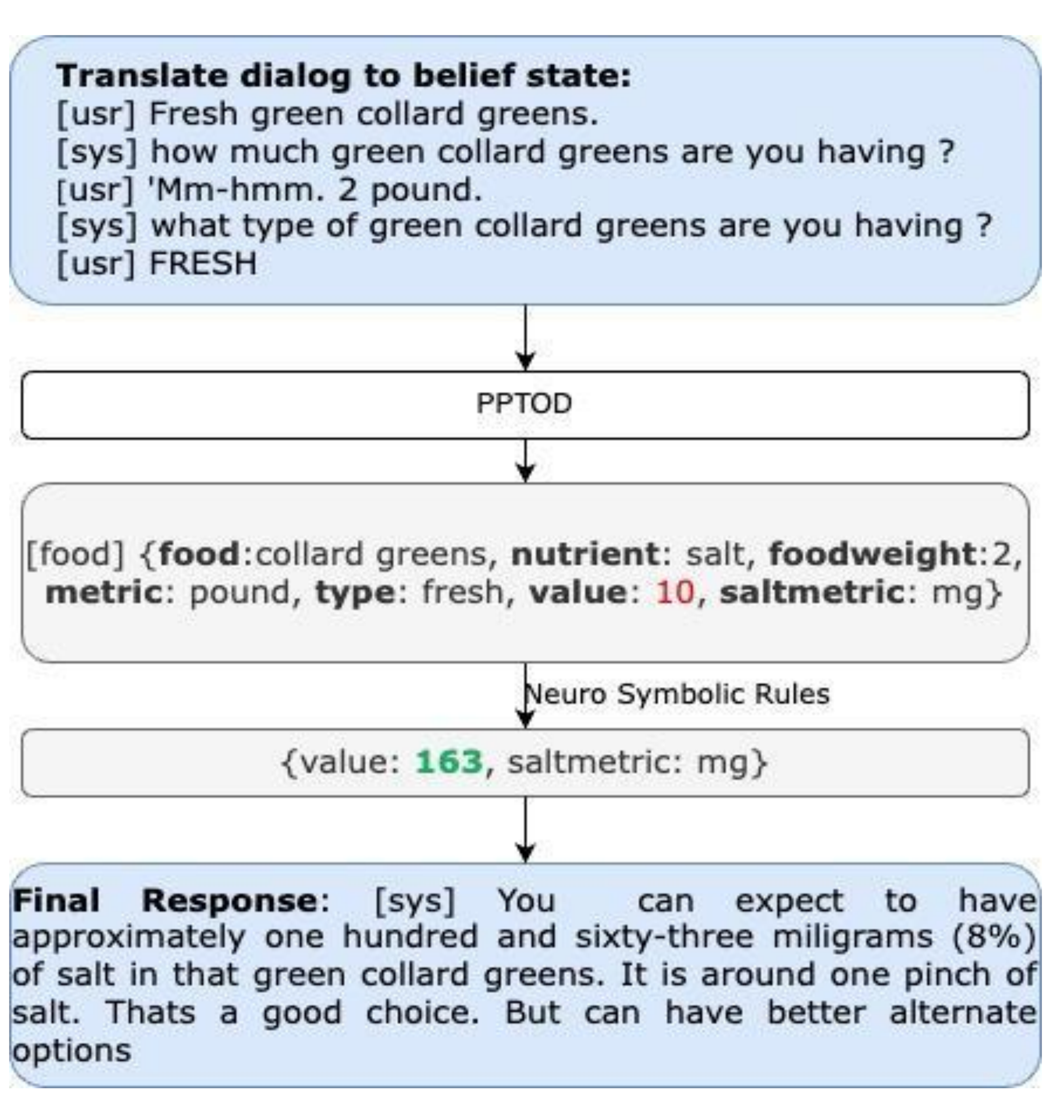
Research Questions

- How to effectively combine strengths of TODS and LMs/LLMs to create a hybrid dialog model?
- How do users/patients/older adults perceive such a system?

Preliminary Work

Need for control :
- LLMs fails following basic instruction[1]
- Prompting can't ensure control/ personalisation
- Unsuitable for patient-centric deployment

Solution:
- Use Neuro-Symbolic approach
- Demonstrate feasibility on Salt Intake TODS [2]



LMs fail at numeric reasoning; Neuro-Symbolic delivers +20% joint accuracy.

	Neuro-Symbolic System	LLM System
Error Analysis	✓	x
Reliable	✓	x
Complex Query	x	✓
Fluent	x	✓
Concise	✓	x
Controlled	✓	x
Create easily	x	✓

AA HF user study: Systems complement each other [3]

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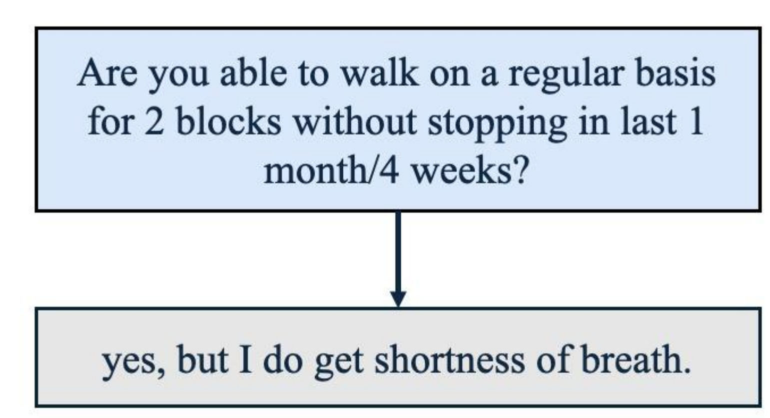
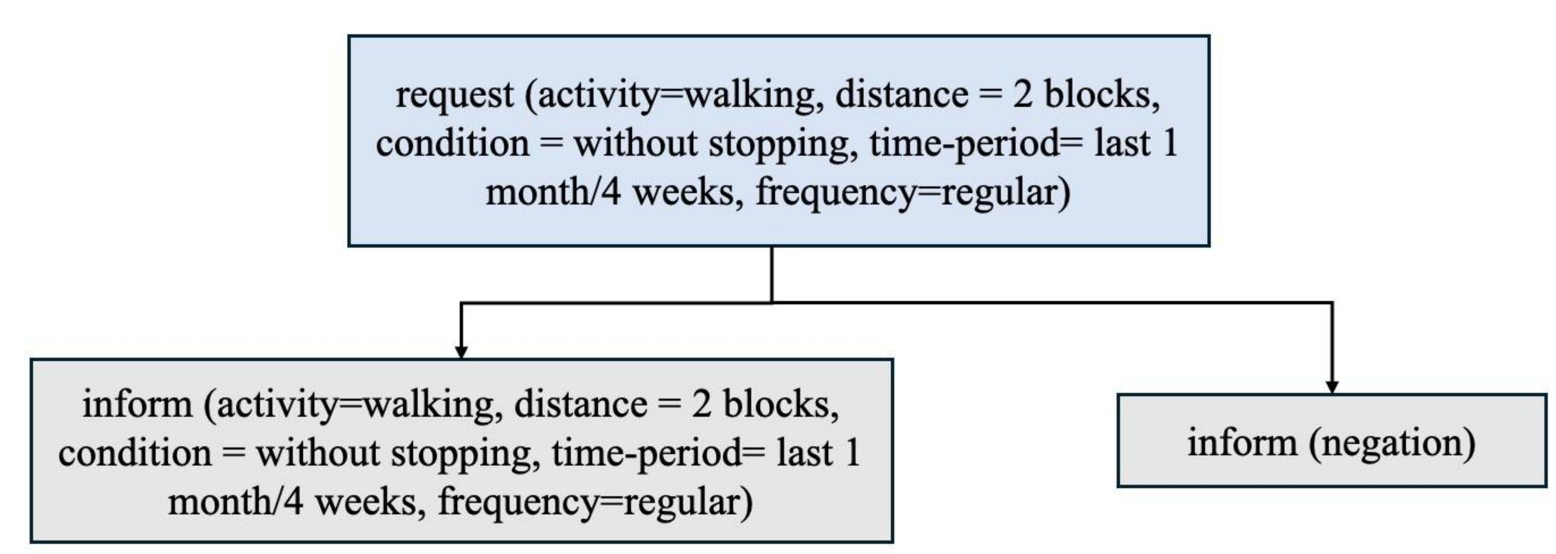
Proposed Work

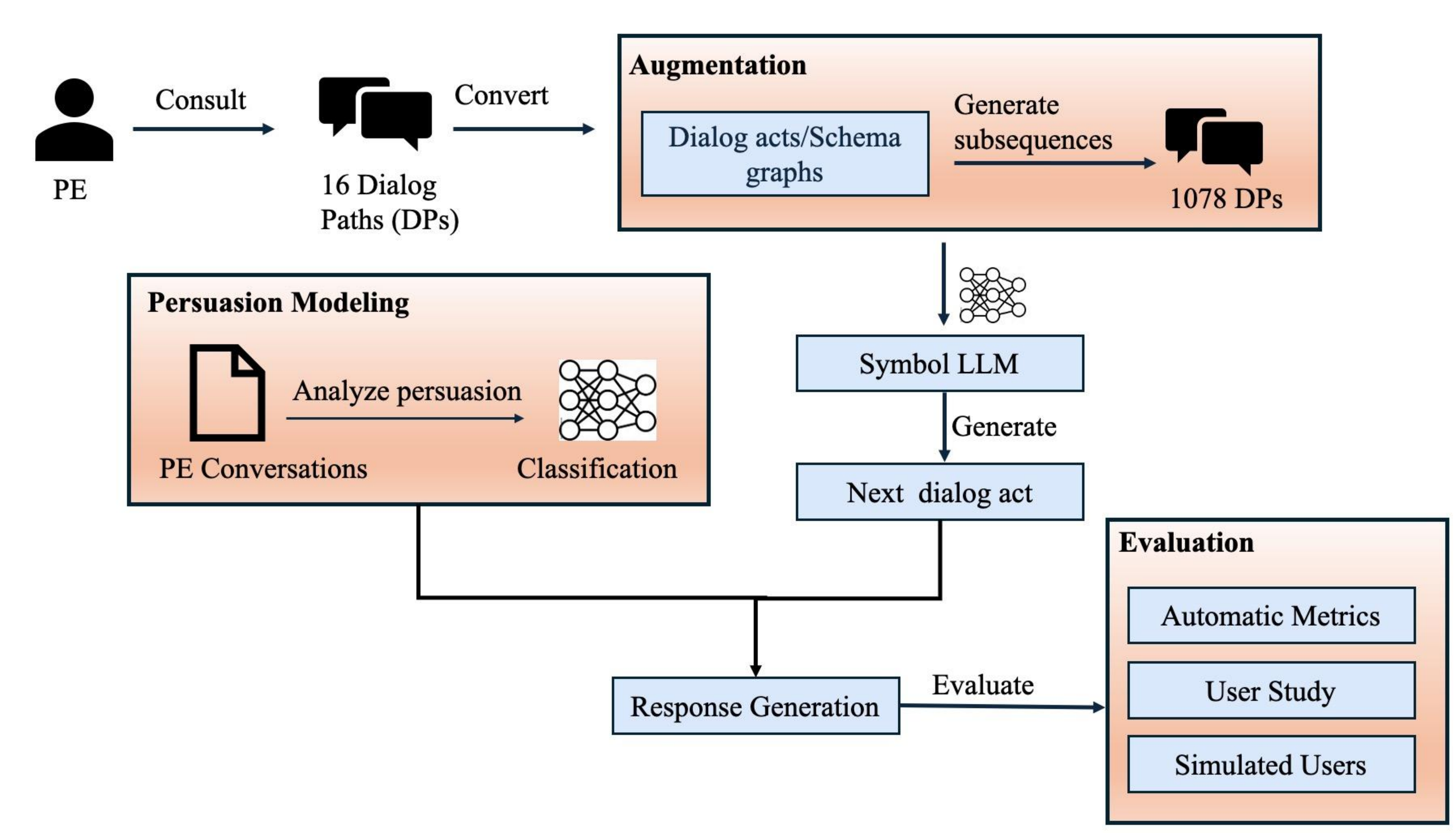
Hypothesis: Controlled dialogs adapted to communication strategy improve patient engagement
- Integrate dialog acts to control conversation flow
- Add persuasion strategies



Dialog Path



Dialog Act Representation of dialog path



Proposed Workflow of Exercise Dialog System

Conclusions/Takeaways

- Create exercise domain conversational dataset.
- Analyze persuasion in Patient-Educator Conversations.
- Demonstrate neuro-symbolic rule embedding for controllable dialog.
- Show feasibility using robust evaluation.

References

[1] A Tayal, D Salunke, B Di Eugenio, P A-Meares, E P Abril, O Garcia, C Dickens, A Boyd. 2025a. 2025b. Towards conversational assistants for health applications: using chatgpt to generate conversations about heart failure. SIGDIAL 25
[2] A Tayal, B Di Eugenio, D Salunke, A D. Boyd, C Dickens, E. Abril, O Garcia, P A-Meares. 2024. A Neuro-Symbolic Approach to Monitoring Salt in Food. In Proceedings of the First Workshop on Patient-Oriented Language Processing (CL4Health) @ LREC-COLING 24, Italia. ELRA
[3] A Tayal, D Salunke, B Di Eugenio, P A-Meares, E P Abril, O Garcia, C Dickens, A Boyd. 2025a. Conversational assistants to support heart failure patients:comparing a neurosymbolic architecture with chatgpt. Submitted (LREC 2026).